



Assessment Cover Sheet 2025/2026

Course Code: UC2SAT102	Course Title: Statistical Analysis Tools & Techniques	Lecturer: Isah Lawal
Assignment No: 1	Total number of pages in this document: 3	Maximum Word Count: NA
Assignment Title: Statistical Data Analysis Course Project I		
Date Set: 20.02.2026	Submission Date: 11:59 Sunday, 01.03.2026	Feedback Date: Within 3 weeks afterward

Section 1: Submission

Record of Submission and Plagiarism Declaration

In submitting completed work for assessment, you declare that this submission is your work and that any collaborative work or reuse of existing material has been explicitly indicated in the text. When submitting completed work, you agree that it may be submitted for plagiarism testing.

Work should be submitted by the above deadline:

- Electronically Via Moodle in the first case. If problems are encountered, then via email to the course staff.
- Ensure that you read and follow all instructions for this assignment and any additional instructions provided by the course staff. Failure to do so may adversely affect your grade.

IT IS YOUR RESPONSIBILITY TO KEEP A COPY OF ALL SUBMITTED WORK.

Section 2: Statistical Data Analysis

Introduction:

In this assessment, you will analyze a real-world dataset from the Canadian rental property market. As a data scientist, your goal is not only to analyze data but also to extract meaningful insights that can assist renters in finding a home or help investors understand market trends.

About the dataset:

The dataset for the assessment contains real estate rental listings in Canada, organized by province and city. The dataset includes over 25,000 property records, each with 18 attributes that provide a mix of numerical (price, beds, sq_foot), categorical (city, furnishing, smoking), and geospatial (lat/long) variables describing each property's characteristics. A CSV file containing the dataset is available on the course page in Moodle for your analysis.

Description of Tasks:

Perform a thorough statistical analysis of the given dataset to generate meaningful insights. Your insights should help to answer the following important questions.

1. What property type offers the best value in terms of the lowest price per square foot? Additionally, how does this ranking differ when comparing furnished versus unfurnished properties?
2. How do the physical dimensions of a property (specifically square footage, number of bedrooms, and number of bathrooms) correlate with rental prices? Among these variables, which one demonstrates the strongest linear relationship with the cost of rent?
3. What is the difference in median rental prices between pet-friendly properties and those that do not allow pets? Additionally, how does this "pet premium" differ across various provinces?
4. How is the distribution of rental prices across different provinces?
5. How does the Furnishing Premium, which represents the additional cost for a furnished unit, vary across different property types? For instance, is the price difference between unfurnished and furnished units greater for a 'Basement' compared to an 'Apartment'?

Reporting of the data analysis:

The report aims to demonstrate that you are proficient at interpreting and presenting the results of your data analysis. Use the LaTeX template provided to report your insights. Emphasis should be placed on professionalism and clarity in the report. The report should be comprehensive and simple enough that a non-technical person can understand. Well-organized and described tables and graphs are good things to have in the report.

Deliverables:

You should submit the following:

- Submit a single PDF file of your report using the submission link provided on the course page in Moodle.
- Submit the code used for your analyses on the same submission link

Grading:

The general breakdown of the grade is as follows:

A: Outstanding effort. The student understands and knows how to choose and apply concepts, can put the analysis into a cogent argument, can identify weaknesses in the analysis, and can communicate the results to others.

B: Good effort. The student understands most of the concepts, puts together an adequate argument, identifies some weaknesses of their analysis, and communicates most results clearly to others.

C: Passing effort. The student has some understanding of concepts but has some trouble putting analysis together in a cogent argument, and the communication of results is somewhat unclear.

D: Struggling effort. The student is making some effort but has some misunderstanding of some of the concepts and is unable to put together a cogent argument. Communication of results is sometimes unclear.

E: Student is not making a sufficient effort. Demonstrates marginal understanding of the concepts and is unable to choose appropriate techniques for a given task. The communication of the results is mostly unclear.

Section 4: Marking and Assessment

This assignment will be marked out of:	A-F
Contribution to the final course mark:	40%
Estimated number of hours to be spent on this assignment:	15hr